



Hệ thống kiểm tra trùng lặp nội dung

KẾT QUẢ KIỂM TRA TRÙNG TÀI LIỆU

THÔNG TIN TÀI LIỆU

Tên tác giả	SuperAdmin
Tên file	NGUYỄN VĂN DUY_20220995_20447_1145.docx
Thời gian nộp	21/05/2025 00:02:52
Thời gian trả kết quả	21/05/2025 00:02:56
Tên file có tỷ lệ tương đồng cao nhất	BÀI KIỂM TRA GIỮA KÌ-BÙI THANH TÙNG.docx
Tỷ lệ tương đồng	89,58%

ĐOẠN VĂN TRÙNG LẶP

```
#include "glut.h"
#include <cmath>
const float PI = 3.14159265f;
// Hàm vẽ hình tròn
void drawCircle(float r, int segments, float R, float G, float B) {
    glColor3f(R, G, B);
    glBegin(GL_POLYGON);
    for (int i = 0; i < segments; i++) {
```

```

float theta = 2.0f * PI * float(i) / float(segments);
float x = r * cosf(theta);
float y = r * sinf(theta);
glVertex2f(x, y);
void drawAxes() {
glColor3f(1.0f, 0.0f, 0.0f); // do
glBegin(GL_LINES);
glVertex2f(-1.0f, 0.0f);
glVertex2f(1.0f, 0.0f);
glColor3f(0.0f, 1.0f, 0.0f); // xanhla
glBegin(GL_LINES);
glVertex2f(0.0f, -1.0f);
glVertex2f(0.0f, 1.0f);
void display() {
glClear(GL_COLOR_BUFFER_BIT);
glLoadIdentity();
drawAxes();
glPushMatrix();
glTranslatef(0.0f, -0.1f, 0.0f);
glColor3f(0.55f, 0.27f, 0.07f);
glBegin(GL_QUADS);
glVertex2f(-0.3f, -0.3f);
glVertex2f(0.3f, -0.3f);
glVertex2f(0.3f, 0.3f);
glVertex2f(-0.3f, 0.3f);
glPopMatrix();
glPushMatrix();
glTranslatef(0.0f, 0.2f, 0.0f);
glBegin(GL_TRIANGLES);
glVertex2f(-0.35f, 0.0f);
glVertex2f(0.35f, 0.0f);
glVertex2f(0.0f, 0.25f);
glPopMatrix();
glPushMatrix();
glTranslatef(0.18f, 0.4f, 0.0f);
glRotatef(-20.0f, 0.0f, 0.0f, 1.0f);
glColor3f(0.36f, 0.25f, 0.20f);

```

```

glBegin(GL_QUADS);
glVertex2f(-0.025f, -0.1f);
glVertex2f(0.025f, -0.1f);
glVertex2f(0.025f, 0.1f);
glVertex2f(-0.025f, 0.1f);
glPopMatrix();
glPushMatrix();
glTranslatef(0.1f, -0.35f, 0.0f);
glColor3f(0.36f, 0.25f, 0.20f);
glBegin(GL_QUADS);
glVertex2f(-0.08f, -0.05f);
glVertex2f(0.07f, -0.05f);
glVertex2f(0.07f, 0.25f);
glVertex2f(-0.07f, 0.25f);
glPopMatrix();
// glPushMatrix();
glTranslatef(-0.18f, 0.0f, 0.0f);
drawCircle(0.05f, 100, 0.0f, 0.5f, 1.0f);
glPopMatrix();
glPushMatrix();
glTranslatef(-0.65f, -0.3f, 0.0f);
glColor3f(0.55f, 0.27f, 0.07f); //maunau
glBegin(GL_TRIANGLES);
glVertex2f(0.0f, 0.0f);
glVertex2f(0.1f, 0.0f);
glVertex2f(0.05f, 0.3f); //venhon
glPopMatrix();
glPushMatrix();
glTranslatef(-0.60f, 0.0f, 0.0f);
drawCircle(0.10f, 100, 0.0f, 0.6f, 0.0f);
glPopMatrix();
void init() {
glClearColor(1.0, 1.0, 1.0, 1.0); // mau trahngt
glMatrixMode(GL_PROJECTION);
gluOrtho2D(-1, 1, -1, 1); // Tọa độ 2D
int main(int argc, char** argv) {
glutInit(&argc, argv);

```

```
glutInitWindowSize(800, 800);  
glutCreateWindow("ve ngoi nha ");  
glutDisplayFunc(display);  
glutMainLoop();
```